
CprE 458/558: Real-Time Systems

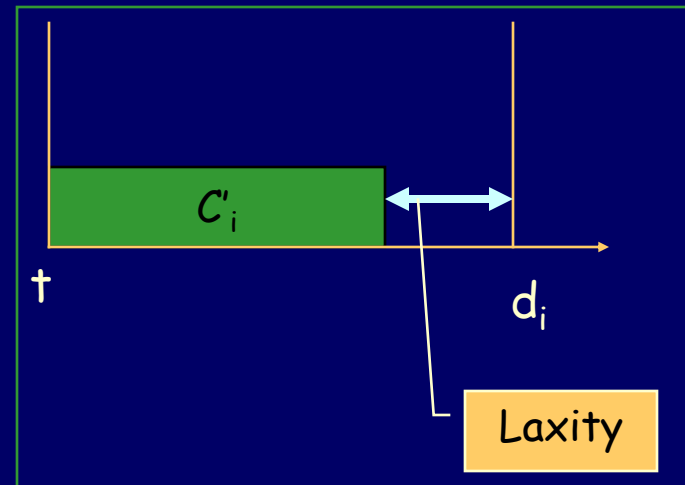
RMS and EDF Schedulers (Priority-driven Preemptive Schedulers)

Priority-driven Preemptive Scheduling

Assumptions & Definitions

- Tasks are periodic
 - No aperiodic or sporadic tasks
 - Job (instance) deadline = end of period
 - No resource constraints
 - Tasks are preemptable
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- Laxity of a Task T_i ,
$$L_i = d_i - (t + c_i')$$

where d_i : deadline;
 t : current time;
 c_i' : remaining computation time.



Rate Monotonic Scheduling (RMS)

- **Schedulability check (off-line)**
 - A set of n tasks is schedulable on a uniprocessor by the RMS algorithm if the processor utilization (utilization test):

$$\sum_{i=1}^n c_i/p_i \leq n(2^{1/n} - 1).$$

The term $n(2^{1/n} - 1)$ approaches $\ln 2$, (≈ 0.69 as $n \rightarrow \infty$).

- This condition is sufficient, but not necessary.

RMS (cont.)

- **Schedule construction (online)**

- Smaller the period higher the priority.

- E.g., Task with the smallest period is assigned the highest priority.

- At any time, the highest priority task is executed.

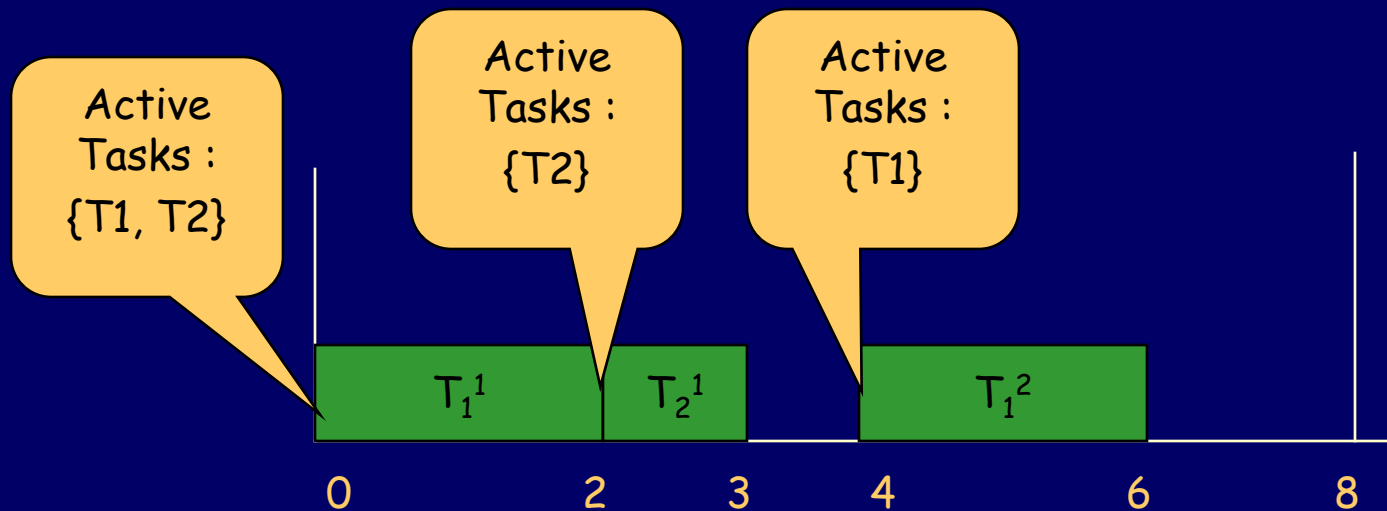
RMS is an optimal preemptive scheduling algorithm with fixed priorities.

Static/fixed priority algorithm assigns the same priority to all the jobs (instances) in each task.

RMS Scheduler -- Example 1

Task set: $T_i = (c_i, p_i)$
 $T1 = (2,4)$ and $T2 = (1,8)$

Schedulability check:
 $2/4 + 1/8 = 0.5 + 0.125 = 0.625 \leq 2(\sqrt{2} - 1) = 0.82$



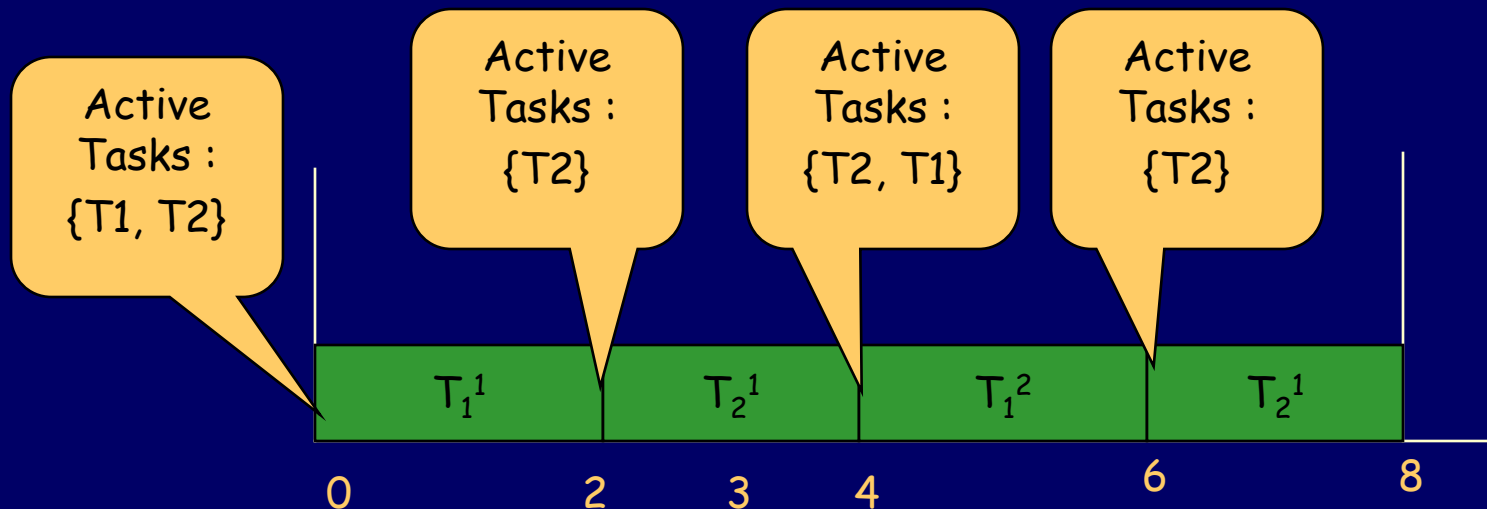
RMS scheduler -- Example-2

Task set: $T_i = (c_i, p_i)$

$T_1 = (2, 4)$ and $T_2 = (4, 8)$

Schedulability check:

$$2/4 + 4/8 = 0.5 + 0.5 = 1.0 > 2(\sqrt{2} - 1) = 0.82$$



Some task sets that FAIL the utilization-based schedulability test are also schedulable under RMS → We need exact analysis (necessary & sufficient)

Earliest Deadline First (EDF)

- **Schedulability check (off-line)**
 - A set of n tasks is schedulable on a uniprocessor by the EDF algorithm if the processor utilization.

$$\sum_{i=1}^n c_i / p_i \leq 1$$

- This condition is both necessary and sufficient.
 - **Least Laxity First (LLF)** algorithm has the same schedulability check.

EDF/LLF (cont.)

- **Schedule construction (online)**
 - EDF/LLF: Smaller the deadline/laxity, higher the priority.
E.g., Task with the smallest deadline/laxity is assigned the highest priority.
 - At any time, the highest priority task is executed.

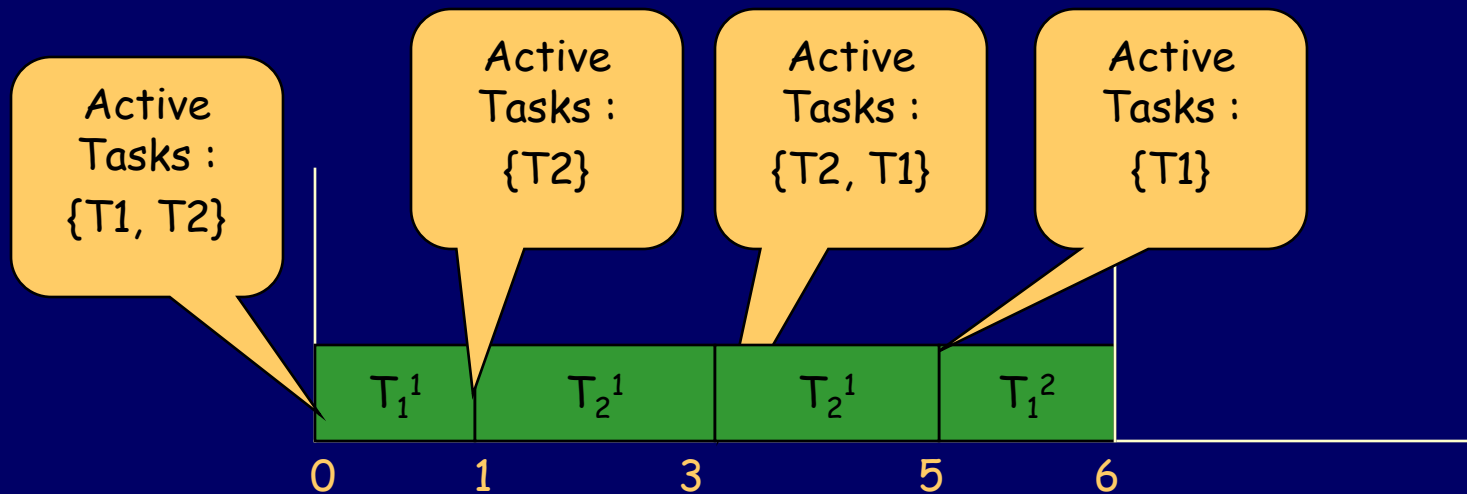
EDF/LLF is an optimal preemptive scheduling algorithm with dynamic priorities.

Dynamic priority algorithm assigns different priorities to the individual jobs (instances) in each task.

EDF scheduler -- Example

Task set: $T_i = (c_i, p_i, d_i)$
 $T1 = (1,3,3)$ and $T2 = (4,6,6)$

Schedulability check:
 $1/3 + 4/6 = 0.33 + 0.67 = 1.0$



Unlike RMS, Only those task sets which pass the schedulability test are schedulable under EDF

RMS vs. EDF/LLF

- RMS is an optimal preemptive scheduling algorithm with fixed priorities.
- EDF/LLF is an optimal preemptive scheduling algorithm with dynamic priorities.
- RMS schedulability properties can be analyzed for complex scenarios; rich theory exists and it is widely used in practice.
- EDF/LLF offers higher schedulability than RMS, but it is more difficult to implement.